

Optimal Queries Processing in a Heterogeneous Sensor Network Using Multicommodity Network Flow Method

Nurjahan Begum¹, Samia Tasnim¹, Mahmuda Naznin¹

¹Department of Computer Science and Engineering,
Bangladesh University of Engineering and Technology, Dhaka – 1000, Bangladesh
(urmee_039@csebuett.org, samia.tasnim@gmail.com, mahmudanaznin@cse.buet.ac.bd)

Abstract - In a wireless sensor network, it is a common practice to have in network partial data processing. As part of this, the mitigation of the demands issued by sink nodes in a heterogeneous resource constrained sensor network requires optimal resource sharing, which is a very challenging task. We find this task analogous to finding the optimal flow in a multi commodity flow network. The minimum cost multi commodity flow problem is an NP hard one. In our paper, we give an optimal cost algorithm to solve this problem where a special data structure is built, which we call Data Forwarding Tree. We compare our algorithm to a well known auction method where a commodity wins the bid based on the lowest cost path, and the other looser commodities are shipped using more expensive paths incurring more cost. An algorithmic analysis shows that our method is free of deadlock as well.

Keywords - Multicommodity Sensor Network, Query Processing, Optimal Algorithm.

I. INTRODUCTION

A wireless sensor network (WSN) is a network consisting of spatially distributed autonomous devices using sensors to cooperatively monitor physical or environmental conditions, such as temperature, sound, vibration, pressure, motion or pollutants, at different locations [1]. These devices have physical constraints like communication, power consumption, computation, uncertainty in sensor readings, etc [2]. Traditional query processing is not applicable to sensor networks due to these constraints involved. A query-processing mechanism for WSNs should be able to handle common conditions such as failures, resource limitations (e.g., energy and memory), the existence of large amounts of data streams and mobility of the sensors [3].

In a WSN, collected data are passed from sensor to sensor until the base station is reached. Traditionally, in sensor networks, routing data structure construction is done by having nodes pick a parent with the most reliable connection to the root (highest *link quality*.) In [4], a special data structure called the *Semantic Routing Tree* is designed to allow each node to efficiently determine if any of the nodes below it will need to participate in a given query over some constant attribute. For bottom to

top data transmission in sensor network, different types of trees have been constructed based on different parameters, as in [5, 6].

Construction of routing data structure by keeping the associated cost minimum is a challenging issue for researchers when the sensor network, along with its physical constraints, involves constraints of traditional multi commodity networks. Different approaches have been adopted for solving the multi commodity minimum cost network flow problem from query processing scenario perspective of WSN as in [7].

In this paper, we formulate our problem domain of construction of a cost balanced distributed data structure which we call *Data Forwarding Tree (DFT)*. We consider the time required for processing a query as the *cost* of data processing in DFT. We develop an algorithm that performs better than a well known auction approach [7]. The paper is organized as follows. Section II describes the problem domain. Section III summarizes our algorithm, Section IV gives the comparative analysis with the auction approach. Section V is the background perspective of our research work. Section VI shows simulation results validating our method and finally, Section VII gives the conclusion with the direction of future extension.

II. SYSTEM MODEL

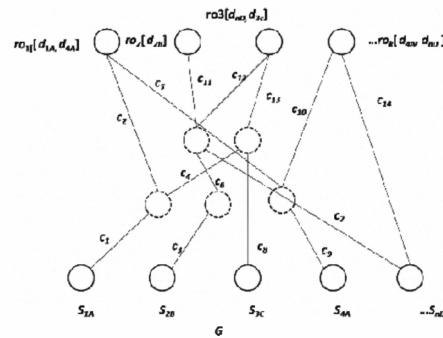


Fig. 1. A Sample Sensor Network

In Figure 1, we show a typical query processing scenario by graph $G(V, E)$ with k target nodes $ro_1, ro_2, \dots, ro_k$ and n source nodes $s_{1A}, s_{2B}, s_{3C}, s_{4A}, \dots, s_{nD}$. Each source node is capable of producing a single type of commodity,

but multiple source nodes can produce same type of commodity. For example, the nodes s_{1A} and s_{4A} produce same commodity A. A target node may demand for one or more type of commodity from one or more source nodes. For example, target node ro_k demands for commodity A and D from source nodes s_{4A} and s_{nD} respectively and this has been shown as d_{4A} and d_{nD} . A target node may demand for the same type of commodity from different source nodes producing the same commodity. For example, ro_1 demands for commodity A from both source nodes s_{1A} and s_{4A} . Neither the source nodes nor the target nodes are connected to themselves. In Figure 1, we show some nodes marked as dashed nodes which we call transshipment centers. These are used to forward data as relay nodes for the passage of data from source nodes to target nodes.

A. Supply, Demand and Capacity

In G , we view each commodity as $K_j = (s_{ij}, ro_i, d_{ij})$, where s_{ij} and ro_i are the source and target node of a commodity j , and d_{ij} is the demand on target node. The amount demanded by a target node ro_i of a specific commodity j from the associated source node s_{ij} is defined as the *demand* d_{ij} and the amount to be supplied of a specific commodity j from the associated source node s_{ij} demanded by a target node ro_i is defined as the *supply* for j . The numerical values of the demand bound and supply bound for commodity j from a specific source to specific target node are equal. The time required for a unit amount of data to be passed via a link $(u, v) \in E$ in $G(V, E)$ is the *capacity* of the link. If the time required for unit amount of data to pass through link (p, q) is less than that of a link (g, h) then, the former is a faster link whereas the later is a slower one.

B. udata and utime

For the advantage of our formulation, we define another two terms. The data from the source node to the target node is passed as multiples of unit data and we call it *udata*, δ . The time involved for the entire demanded data to reach to the corresponding target node can be thought of as multiples of unit time and we define it as *utime*, α .

Now, we give an assumption which is required to define the cost model of our network flow model.

Assumption 1. *There can never be simultaneous existence of first occurrences of two udata in the same time scale.*

For Assumption 1, between the first occurrences of any two consecutive δ , a delay time has to be introduced. We call this delay *dtime* and denote it by τ .

C. Cost Metric

The cost metric of our research is the time required for the fulfillment of the demands for all of the

commodities requested by all of the target nodes. We define, final cost incurred,

$$c_F = \max\{ \text{for commodity } j \{ \mu_j \alpha_j \} \} \quad [\text{for satisfying demand for commodity } j, \eta_j \delta_j = d_{ij}]. \quad (1)$$

In (1), $j = 1$ to r as requested by the target nodes;

$i = 1$ to n stands for respective supplier node;

μ_j and η_j are integer multiples of α and δ respectively corresponding to commodity j .

The objective of this research paper is to propose an algorithm by which under the resource constrained environment as stated above, the demand requirement of all the target nodes may be accomplished by keeping the cost involved minimum. In our algorithm proposed, we determine the formal model for the determination of the size of the time window under some special cases by which the incurred cost can be calculated.

D. Construction of Data Forwarding Tree

For each commodity demanded by the target nodes, we find the shortest path from source to sink nodes. The shortest path is determined by Dijkstra's algorithm. After finding all paths, we merge them and form our special data structure called *Data Forwarding Tree (DFT)*. We construct this data structure by merging the unique shortest paths from the same target node to multiple associated source nodes which results in a tree. Eventually if a source node has multiple target nodes for supplying data to, then it will be member of multiple DFTs accordingly.

In the constructed DFT, we define four types of links –

- Type A: Source node to transshipment node
- Type B: Source node to target node
- Type C: Transshipment node to target node
- Type D: Transshipment node to transshipment node

Lemma 1. *If the cost of the link connecting two nodes A (source node) and B (target node or distribution center) is c , i.e if the link is either a Type A or Type B link of a DFT, and the demand of the associated commodity is η , then the cost incurred for full satisfaction of the demand from A to B is $c + (\eta - 1)$.*

Proof (By induction)

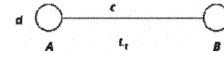


Fig. 2. A Sample Link (Type A or Type B)

Let us consider the Figure 2. Node A is producing δ and this data is transmitting through link L_1 . This link has a cost of c and the demand issued by node B is d . We consider for full satisfaction of the demand, $d = 1, 2, \dots, \eta$ units of δ needs to be passed.

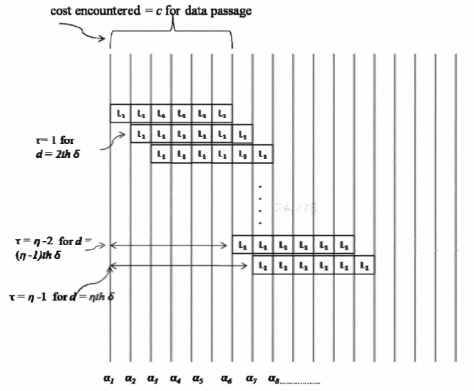


Fig. 3. Cost encountered while δ transmission

(Basis Step): For $d = 1$, that is, when the first δ is transmitted through L_1 it will not encounter any delay time. Therefore, cost encountered for data transmission = c and cost encountered for delay time = $\eta - 1 = 1 - 1 = 0$. Therefore, final cost, $c_{F1} = \text{cost for data transmission} + \text{cost for delay time} = c + (1 - 1) = c$.

(Induction hypothesis): Let, for $d = \eta - 1$, the statement to be proved holds.

That means, $c_{F\eta-1} = c + (\eta - 1 - 1)$
 $= c + (\eta - 2)$

(Induction step): We need to prove for $d = \eta$, the statement to be proved holds as well. That means, we need to prove $c_{F\eta} = c + (\eta - 1)$ holds.

$$\begin{aligned} \text{Now, } c_{F\eta} &= c + (\eta - 2 + 1) \\ &= c + (\eta - 2) + 1 \\ &= \pi + 1 \text{ [considering } \pi = \eta - 2] \end{aligned} \quad (2)$$

The quantity marked with π holds for $d = \eta - 1$ (from induction step). Therefore equation 2 holds for $d = \eta$. (Proved)

Lemma 2. When a link is shared by multiple δ (from same or different source nodes), then the assumption of the existence of not more than one instance of the first occurrence of any δ along that link is sufficient for the calculation of delay time encountered during the passage of δ along that link.

Proof

Let us consider Figure 3. When the first δ ($d = 1$) starts transmitting first along the link L_1 , let the associated timeline be α_1 . The second ($d = 2$) δ starts transmitting at timeline $\alpha_2 = \alpha_1 + (2 - 1)$. Therefore, for the η th δ , the starting timeline would be $\alpha_\eta = \alpha_1 + (\eta - 1)$.

Therefore, when multiple δ passes through a shared link, all of them cannot start during the same timeline. A specific occurrence of a specific δ occupies a specific position of the shared link. No two occurrences of different δ can occupy the link during the same timeline. The startup delay introduced therefore ultimately reflects the encountered final cost accordingly as proved in Lemma 1.

It is therefore sufficient to assume not more than one instance of the first occurrence of any δ along the shared link will exist and this will introduce the overall offset

delay along that link. For this there will be no chance of conflict of any specific occurrence of any two δ passing through that link during the same timeline. (Proved)

Lemma 3. There is no loss of generality in removing a transshipment node with one parent and one child in a DFT by connecting the parent and the child via a single link with the cost assigned as sum of the costs of the links connecting the child and the transshipment node and the transshipment node and the parent.

Proof

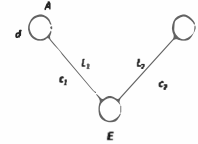


Fig. 4(a). Sensor network with transshipment node

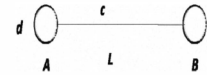


Fig. 4(b). Sensor network with transshipment node removed

Let us consider the two figures above. In Figure 4 (a), we show a sensor network with a transshipment node E. Node A can be either source node or distribution center and node B can be either target node or distribution center. Therefore, L_1 may be a Type A or Type D link and L_2 may be a Type C or Type D link. The link connecting node A and E has cost c_1 and the one connecting node E and B has cost c_2 . The demand for commodity coming from node A is d .

According to lemma 2, we need to consider the delay incurred for cases like above only once. Therefore, the delay encountered for the network in Figure 4 (a) is $(d - 1)$.

$$\text{Final cost, } c_{F1} = c_1 + c_2 + (d - 1) \text{ [from lemma 1]} \quad (3)$$

Now let us consider Figure 4 (b). This is the version of the network in Figure 4 (a) with transshipment node E removed. A and B are connected directly with link L having cost $c = c_1 + c_2$

$$\text{Therefore, final cost, } c_{F2} = c + (d - 1) \text{ [from lemma 1]} \quad (4)$$

$$= c_1 + c_2 + (d - 1) \text{ [from equation 4]} \quad (5)$$

$$\text{From equation 3 and 5, we find, } c_{F1} = c_{F2}$$

Therefore the statement to be proved holds. (Proved)

Using the property described in lemma 3, we call the operation performed *link normalization*.

E. Waiting Time Calculation

$$F_{1p} = \begin{cases} 1, & \text{if } p\text{th source is the first source node the first } \delta \\ & \text{of which has been allowed to transmit through} \\ & \text{the shared link} \\ 0, & \text{otherwise} \end{cases}$$

$$F_{2p,q} = \begin{cases} 1, & \text{if the previous } \delta \text{ of } p\text{th source's } q\text{th } \delta \text{ passed} \\ & \text{through the shared link is of a different source} \\ & \text{node} \\ 0, & \text{otherwise} \end{cases}$$

$w_{p,q}$ = waiting time of the previous δ of p th source node's q th δ transmitted via the shared link

Therefore, waiting time for p th source node's q th $\delta =$

$$W_{p,q}^{wait} = \begin{cases} w_{p,q}, & \text{if } F_{1p} = 1 \text{ and } F_{2p,q} = 1 \\ w_{p,q} + 1, & \text{otherwise} \end{cases}$$

F. Final Cost Equation

Let, S = number of source nodes

$W_{p,q}^{wait}$ = waiting time for p th source nodes's q th δ

σ = number of shared links

c_{sh}^r = cost of r th shared link

ρ_{sh} = sum of the cost of the shared links $= \sum_{i=1}^{\sigma} c_{sh}^i$

c_{β}^k = k th link cost of β th source node

d_{β} = demand of δ from β th source node

$g_{\beta} = \begin{cases} 1, & \text{for the source node with } \max(c_{1\beta} + d_{\beta} - 1) \\ 0, & \text{otherwise} \end{cases}$

Therefore, final cost equation will be,

$$C_F = \text{for } j = 1 \text{ to } S \{ \{ \max(c_{1j} + d_j - 1) \} + (\text{for } g_j = 1) W_{j,d_j}^{wait} \} + \rho_{sh} \quad (7)$$

For the case when paths encounter which are to be shared among different groups of commodities, we will let the group of commodities pass in a round robin fashion, i.e. one group will get chance to use the shared path at a time. Only one representative will be selected from a commodity group and this will again be done in round robin fashion.

III. ALGORITHM

Algorithm *costFinder* (S, T, D)

S : Collection of source nodes producing different commodities

T : Collection of Target Nodes issuing demand for different commodities

D : Collection of demands for different commodities

L : current link collection for the current level of DFT

i : sequence number of the associated commodity

p : sequence of source nodes after prioritization

begin

1. $Tr = \text{construct_DFT}(S, T)$ /* Tr is the DFT*/

2. $\text{normalize_link}(Tr)$

3. $\text{level} = 0$ /* level : current level of the Tr */

4. for each level of Tr

$L = \text{traverse_bottom_up}(Tr, \text{level})$

for the returned links in L

i. if the current link is not shared among multiple commodities

$\tau_i = \text{waiting_time_function}(L, i)$

cost of $\delta_i = \alpha_i + \tau_i + d_i - 1$

else if the current link is shared among multiple commodities

p = perform prioritization

do round robin scheduling on the

prioritized commodity sequence p

by letting the winner commodity's δ

to go first and then the one of the loser

$\tau_i = \text{waiting_time_function}(L, i, p)$

cost of $\delta_i = \alpha_i + \tau_i + d_i - 1$

end

IV. COMPARATIVE ANALYSIS

A. Deadlock Avoidance

Lemma 6. Given a multi commodity query processing graph of sensor nodes $G(V, E)$ with n source nodes ($n > 1$), if there is only one path from the target node to the respective source nodes and if the paths from the target node to the source nodes all overlap in such a way that there lies at least one common link for all possible source node pairs, then the auction approach will be deadlocked, but the proposed heuristic will reach convergence.

Proof

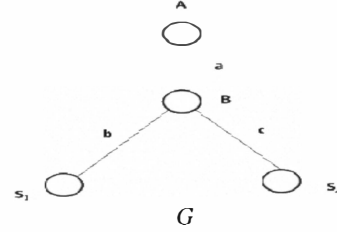


Fig. 5. Problem of Auction Approach (Infinite Bid Value)

Let us consider the query processing graph of sensor nodes, G in Figure 5. Node A is the target node and S_1 and S_2 are the source nodes.

Case 1: (Auction approach) Both the source nodes have only one path towards A and therefore, according to the auction approach, bids infinity. Hence, no commodity wins auction and this process gives rise of deadlock (Proved).

Case 2: (Proposed heuristic) In Figure 5, the link to be shared by both commodities is $B-A$. The two commodities share the link according to the weights assigned there. Eventually, both commodities can satisfy the demand issued by A and the process will converge thereby (Proved).

Lemma 7. Given a multicommodity query processing graph of sensor nodes $G(V, E)$ with n source nodes ($n > 1$), if there is more than one path from the target node to the respective source nodes, if the number of paths possible from the target node to the source nodes is same for all source nodes, if the difference of all consecutive shortest path costs is same for all source nodes and if for a particular source node's path, there is at least one link sharing for all possible source node pairs consisting this node and any other source node, then the auction approach will be deadlocked, but the proposed heuristic will reach convergence.

Proof

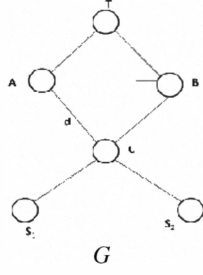


Fig. 6. Problem of Auction Approach
(Same Bid Value)

Let us consider the query processing graph of sensor nodes, G in Figure 6. Node T is the target node and S_1 and S_2 are the source nodes. Let, the paths from S_1 to T are, $S_1 - C - A - T$, $S_1 - C - B - T$, $S_1 - C - A - B - T$ and $S_1 - C - B - A - T$ having length a , b , c , d respectively. The paths from S_2 to T are, $S_2 - C - A - T$, $S_2 - C - B - T$, $S_2 - C - A - B - T$ and $S_2 - C - B - A - T$ having length e , f , g , h respectively. Now, let,

$$\begin{aligned} b - a &= f - e \\ c - b &= g - f \\ d - c &= h - g \\ d - a &= h - e \end{aligned}$$

Case 1: (Auction approach) Here the differences of all consecutive shortest paths are same. This means, when there will be auction for common link usage, then both source nodes will bid with the same value. As a result, auction will result in a draw, and the commodities have to reroute. While rerouting, the commodities eventually have to bid for the common link again, and due to bidding with same bid value, same result continues forever, resulting in deadlock (Proved).

Case 2: (Proposed heuristic) According to the heuristic proposed in this paper, a tree data structure is developed and when there is competition for common link, its usage is distributed among the competitors. Therefore, there lies no chance of deadlock and the algorithm converges (Proved).

V. RELATED WORKS

For solving classical assignment problems like the minimum cost network flow problem, an auction algorithm has been proposed in [7]. Auction approaches have internally economic interpretation intuition and are well suited for implementation in parallel computing systems [9].

Sensor networks dealing with heterogeneous types of data can be structured as multicommodity flow problem. Solving the minimum transportation cost, satisfying all demands, with unsplittable commodities in a multicommodity network flow model is proved to be NP-hard [10]. But, if the commodities are splittable, then the problem is solvable in polynomial time.

In [8], the authors have given a solution to multi commodity flow problem from sensor network perspective and they prove the solution to be optimal as well as distributed. But they have not considered weighted

links for which their algorithm will not always return minimum cost.

In [11], the authors have proposed a tree based routing protocol that focuses on prolongation of sensor network lifetime, but the time required for total demand satisfaction has not been considered here.

VI. SIMULATION RESULTS

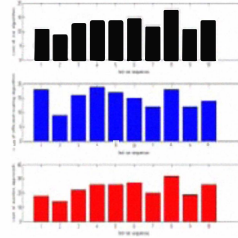


Fig. 7. Comparative performance
of the three algorithms

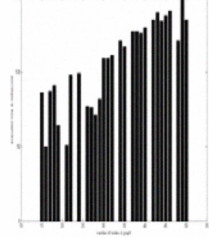


Fig. 8. Execution time of the
proposed algorithm

We have simulated the performance of our proposed algorithm, the one with optimal efficient routing and the one depicting the auction approach in MATLAB simulator and in Figure 7, we show the comparative performance of these three algorithms for ten test runs. We find that when the hop count of the shortest path from the source node to the target node is least, then the efficient routing algorithm performs same as our proposed algorithm. For all other cases, our algorithm gives lower cost than the auction approach and the efficient routing approach. Figure 8 shows the change in execution time of our proposed algorithm with respect to change in number of nodes in the sensor network.

VII. FUTURE WORK AND CONCLUSIONS

In our paper, we find that time efficient multiple queries processing in a heterogeneous sensor network analogous to finding the minimum cost multiple commodities flow in a network. We give an efficient solution to solve the minimum cost multicommodity network flow problem. We compare our algorithm to two other minimum cost multicommodity network flow solution methods. We also prove that our method is distributed and free of deadlock. In future, we would like to incorporate other network properties like signal strength, security levels, mobility in our model.

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